PRISMATIC ANALYSIS: THE GLOBAL RESPONSE TO ANTIMICROBIAL RESISTANCE

A Mark 2 Application of the Prismatic Method

Case Study: The Global Response to Antimicrobial Resistance (AMR), 2015–2026 **Method:** The Prismatic Method (R. Jazinski, *Silent Transitions*, 2026) **Analyst:** Claude (Anthropic), operating under explicit methodological instruction **Date:** March 2026

STEP 0: PRE-ANALYSIS REFLEXIVE DECLARATION

i. What is my likely "home grammar" for this topic?

My likely home grammar is what I will call the **Biomedical-Institutional Grammar** — a framework that treats AMR primarily as a clinical and public health problem requiring coordinated technocratic intervention. This is the grammar of the WHO Global Action Plan, the Lancet GRAM studies, and the infectious disease community. I am drawn to it because it foregrounds quantitative evidence, systematic reviews, and institutional coordination — all of which align with how I process information and construct authority claims.

ii. Which grammar will be hardest for me to operate faithfully? Why?

The grammar I anticipate finding hardest to inhabit is what I will call the **Political Economy Grammar** — a framework that treats AMR as primarily a market failure and governance problem embedded in global capitalist structures. Operating this grammar faithfully requires treating the microbiological evidence as subordinate to structural economic analysis, which cuts against my tendency to privilege the empirical data as foundational. I will need to resist translating its claims back into biomedical terms.

iii. What might I miss given my analytical position?

I am likely to underweight the degree to which the AMR problem is constituted differently by different grammars — that is, I may treat the biomedical framing as the "real" problem and the others as derivative perspectives on it. I may also miss the force of arguments from the ecological grammar that refuse to centre the human clinical encounter at all.

iv. What is my temporal orientation?

I am naturally drawn to causal explanation (backward-looking) and prescription (forward-looking). I am less comfortable with hermeneutic or meaning-making analysis — with asking what AMR *signifies* about modernity's relationship to biological systems rather than what should be *done* about it.

SECTION 1: OBSERVATION FIELD DECLARATION

The Phenomenon

The observation field is the global response to antimicrobial resistance from the adoption of the WHO Global Action Plan on AMR (2015) to March 2026. The geographic scope is global, though the analysis necessarily foregrounds regions where data infrastructure exists (Europe, North America, parts of East and South Asia) and must reckon with the epistemic silence from regions where surveillance capacity is weakest — precisely the regions where AMR burden is highest.

Why This Phenomenon Produces Interpretive Divergence

AMR is often assumed to be a domain of consensus: bacteria evolve resistance to antibiotics, this is bad, and something must be done. This apparent consensus conceals deep structural disagreements about *what kind of problem* AMR is. The microbiological facts are not in serious dispute. What diverges is the causal architecture — whether the primary driver is biological evolution, market failure, agricultural practice, colonial inequity, or ecological disruption — and consequently, what constitutes an adequate response. The decade-long gap between the WHO Global Action Plan (2015) and the first operational subscription-model contracts (2022–2024) is itself a signal of this divergence: the institutional machinery moved slowly not because the biology was uncertain but because competing frameworks produced incompatible intervention logics.

Signal Contestation Level

This is a *mixed-signal domain*. Most core epidemiological signals (mortality estimates, resistance rates, pipeline counts) are shared across grammars, though they are weighted very differently. However, several signals are genuinely contested: whether agricultural antibiotic use is a primary or secondary driver of human clinical resistance; whether the antibiotic pipeline crisis is a scientific problem or an economic one; and whether One Health surveillance data constitute evidence of ecological interconnection or merely correlated trends. The contested signals are identified in Section 2.

Major Interpretive Communities

Three principal interpretive communities occupy this field: (a) the biomedical-institutional community (WHO, CDC, ECDC, infectious disease clinicians, microbiologists, epidemiologists); (b) the political economy community (health economists, development economists, global health governance scholars, the pharmaceutical industry itself in its public-facing advocacy); and (c) the ecological-systems community (One Health advocates, veterinary public health, environmental microbiologists, planetary health scholars, and increasingly, STS scholars treating AMR as a multispecies problem). These communities are not sealed — individuals move between them, and hybrid positions exist — but they do operate distinct interpretive architectures.

SECTION 2: SIGNAL SET

Core Signals

S1. Global AMR mortality estimates (GRAM 2019/2024) *Source:* Lancet GRAM Project. *Type:* Modelled epidemiological estimate. *Data:* 1.27 million deaths directly attributable to AMR and 4.95 million associated with AMR in 2019. The 2024 GRAM update projects up to 1.91 million direct AMR deaths per year by 2050, with 39 million cumulative deaths possible between 2025 and 2050. *Ambiguity:* The counterfactual structure (what would have happened if infections were drug-susceptible vs. no infection) creates two very different numbers. Grammars select between them. The "associated" figure is nearly four times the "attributable" figure. Which number is cited reveals the grammar in operation.

S2. Antibiotic clinical pipeline data *Source:* WHO pipeline reviews (2017–2025). *Type:* Inventory data. *Data:* As of the 2025 WHO review, only 32 antibiotics in clinical development targeting priority pathogens, with only 12 classed as innovative and just 4 targeting critical-priority organisms. Between 2017 and 2023, 13 new antibiotics received market approval globally. Four new antibiotics approved in 2024; gepotidacin (first-in-class) approved 2025. *Ambiguity:* Whether this pipeline is "thin" depends on comparison class. Against oncology (1,800+ immuno-oncology drugs in development), it is catastrophically thin. Against the historical baseline of antibiotic classes (none truly novel since the 1980s), the recent first-in-class approvals represent progress.

S3. Pharmaceutical industry disinvestment *Source:* Industry reports, WHO/G7 documentation. *Type:* Market behaviour data. *Data:* Antibiotic development is nearly absent from the 2025 pipelines of the 20 largest pharmaceutical companies. In 2020, the industry raised £5.4 billion for oncology R&D but only £125 million for antibiotics. Several antibiotic-focused companies have gone bankrupt despite holding approved products. Experts are exiting AMR R&D. *Ambiguity:* Is this evidence of market failure (the Political Economy Grammar) or evidence of inadequate scientific innovation making investment unattractive (Biomedical Grammar)? The same departure pattern supports different causal stories.

S4. Agricultural antibiotic consumption trends *Source:* Van Boeckel et al. (PLOS Global Public Health, 2023); Nature Communications (2025). *Type:* Modelled consumption estimate. *Data:* Global baseline of approximately 110,777 tonnes of veterinary antimicrobials in 2019. Under business-as-usual, projected to reach approximately 143,481 tonnes by 2040 (29.5% increase). Asia projected to account for roughly 63% of global livestock antibiotic use by 2040. Global governments have committed to 30–50% reduction by 2030. *Ambiguity:* The gap between commitment (30–50% reduction) and projection (29.5% increase) is itself a signal whose meaning is grammar-dependent.

S5. Colistin as tracer signal *Source:* Multiple (Lancet Infectious Diseases, CIDRAP, PMC reviews). *Type:* Case-study signal. *Data:* China banned colistin as a feed additive in 2016/2017. Subsequent data showed declining mcr-1 prevalence in China. However, China exported 2,664 tonnes of colistin to 21 countries between 2018 and 2021. Bangladesh and Nigeria imported colistin for livestock despite national bans. MCR-1 gene found in 7% of samples in a Pakistani study, above the global average. *Ambiguity:* Colistin is a signal that every grammar reads differently. For the Biomedical Grammar, it is a story about resistance gene transmission (mcr-1). For the Political Economy Grammar, it is a story about regulatory arbitrage in global trade. For the Ecological Grammar, it demonstrates the futility of single-pathogen, single-country interventions in an interconnected biosphere.

S6. UK subscription ("Netflix") model *Source:* NICE, NHS England, UK Parliament. *Type:* Policy innovation signal. *Data:* Pilot launched 2019, contracts awarded to Pfizer (Zavicefta) and Shionogi (Fetcroja) at £10 million per year each. Model permanently adopted May 2024. Expansion tendered August 2024 at estimated £1.9 billion over 16 years, with payment bands of £5–20 million per year. First operational subscription model in the world. *Ambiguity:* Whether this represents a breakthrough (delinking payment from volume) or a sticking plaster (payments too small to incentivise genuine pipeline investment globally) is sharply contested.

S7. WHO Global Action Plan adoption and NAP implementation *Source:* WHO Library of National Action Plans. *Type:* Governance data. *Data:* 178 countries had developed NAPs by 2024. Implementation rates vary dramatically. Many NAPs lack dedicated funding. *Ambiguity:* Whether plan adoption without implementation constitutes progress or performative compliance is grammar-dependent.

S8. AMR mortality age distribution shift *Source:* GRAM 2024 update. *Type:* Demographic-epidemiological data. *Data:* AMR deaths in children under 5 declined by 50% between 1990 and 2021. Deaths in those over 70 increased by over 80%. By 2050, approximately 66% of attributable AMR deaths will be in those over 70. *Ambiguity:* Shared signal, but the explanation differs. For the Biomedical Grammar, this reflects improved paediatric infection control. For the Ecological Grammar, it reflects demographic transition and comorbidity burden.

S9. Surveillance data quality gap *Source:* GLASS (WHO Global Antimicrobial Resistance and Use Surveillance System). *Type:* Meta-data signal. *Data:* Persistent data gaps in low-income settings. Sub-Saharan Africa, which bears the highest per-capita AMR mortality burden, has the weakest laboratory and surveillance infrastructure. *Ambiguity:* Is surveillance a precondition for action (Biomedical), a governance challenge (Political Economy), or irrelevant to the ecological dynamics already in motion (Ecological)?

S10. OECD/WHO economic modelling *Source:* OECD, World Bank, O'Neill Review (2016). *Type:* Modelled economic projection. *Data:* The O'Neill Review estimated AMR could cost $100 trillion and 10 million lives per year by 2050. OECD modelling projects significant GDP losses. The GRAM 2024 "better care" scenario estimates 92 million cumulative deaths averted between 2025 and 2050 through improved infection management and antibiotic access. *Ambiguity:* Economic projections are highly model-dependent. The 10 million figure has been criticised as methodologically inflated.

S11. EU ban on growth-promoter antibiotics (2006) and regulatory expansion *Source:* EU legislation, EU Regulation 2019/6, EU 2022/1255. *Type:* Regulatory signal. *Data:* The EU banned growth-promoter antibiotics in 2006. The 2022 implementing regulation designated certain antimicrobials as reserved for human use only. Corresponding declines in agricultural antibiotic use and some resistance markers in member states. *Ambiguity:* Whether EU regulatory success is generalisable to LMICs with different agricultural structures is contested.

S12. One Health framework adoption *Source:* WHO/FAO/WOAH Quadripartite. *Type:* Governance-conceptual signal. *Data:* One Health formally adopted as the governance framework for AMR by WHO, FAO, WOAH, and UNEP. The 2024 UN High-Level Meeting declaration endorsed this framing. *Ambiguity:* Whether One Health is an operational framework or a rhetorical umbrella that papers over unresolved jurisdictional conflicts is itself a contested question.

Counter-Signals

CS1. China's colistin ban success. Following the 2017 ban, studies showed declining mcr-1 prevalence in Chinese livestock and human populations. This counter-signal challenges the Political Economy Grammar's scepticism about state regulatory capacity and the Ecological Grammar's pessimism about containment.

CS2. Under-5 AMR mortality decline. The 50% reduction in paediatric AMR deaths since 1990 demonstrates that targeted intervention works. This challenges any grammar that implies the problem is intractable.

CS3. The 2024–2025 first-in-class approvals. Gepotidacin and the Roche zosurabalpin programme suggest the innovation bottleneck may not be permanent. This challenges the Political Economy Grammar's market-failure thesis in its strongest form.

CS4. Norwegian salmon aquaculture. Norway eliminated antibiotics from salmon farming in the 1990s while growing the industry. This demonstrates that antibiotic-free production is economically viable in at least some contexts.

CS5. Persistence of AMR in antibiotic-free environments. Resistance genes persist in environments long after antibiotic pressure is removed. This challenges any grammar premised on the idea that reducing use will straightforwardly reduce resistance.

Contested Signals

Contested Signal 1: Agricultural-to-human resistance transmission

- *Biomedical Grammar reads this as:* a real but secondary transmission pathway compared to hospital and community overuse. The causal evidence is strongest for specific pathogen-drug combinations (colistin/mcr-1) but weaker for aggregate resistance burden.
- *Political Economy Grammar reads this as:* a regulatory failure driven by agribusiness interests that externalise costs onto public health systems.
- *Ecological Grammar reads this as:* the central demonstration that resistance operates as an ecological phenomenon across species and environments, making the human/animal/environmental distinction artificial.

Contested Signal 2: The "broken market" narrative

- *Biomedical Grammar reads this as:* background context — the market is broken, but the primary problem is still the biological challenge of finding new mechanisms of action.
- *Political Economy Grammar reads this as:* the core signal — the market failure *is* the problem, and all downstream consequences (thin pipeline, surveillance gaps, access inequity) flow from it.
- *Ecological Grammar reads this as:* symptomatic of the deeper error — the assumption that the solution to an ecological problem is a pharmaceutical product.

SECTION 3: GRAMMAR SPECIFICATIONS

Grammar A: The Biomedical-Institutional Grammar

Source Tradition: Clinical microbiology, infectious disease medicine, epidemiology, public health institutions (WHO, CDC, ECDC). The Lancet GRAM project is its flagship knowledge infrastructure.

Required Parameters:

1. **Primary analytical variable:** Pathogen-drug resistance as a measurable biological phenomenon, quantified through surveillance, laboratory data, and epidemiological modelling.
2. **Signal selection rules:** Admits signals that can be quantified through clinical or microbiological measurement. Privileges systematic reviews, randomised controlled trials, and modelled estimates grounded in isolate-level data. Requires evidence hierarchies.
3. **Weighting priority:** Mortality data and clinical outcome data are weighted highest. Pipeline data is weighted as a secondary response indicator. Agricultural data is admitted but weighted below clinical data.
4. **Causal chain type:** Linear-proximate. Resistance is caused by selective pressure from antibiotic exposure. Exposure occurs through prescription (human) and administration (animal). Interventions target the proximate cause: stewardship, infection prevention, new drug development.
5. **Admissible closure:** The problem is solvable through evidence-based intervention: improved stewardship, expanded surveillance, strengthened infection prevention, and a robust drug pipeline. Closure takes the form of "if we do X, mortality will decrease by Y."

Recommended Parameters:

6. **Excluded signals:** Tends to exclude or underweight structural economic analysis, geopolitical power dynamics, colonial legacies in health infrastructure, and the environmental resistome (resistance genes in non-clinical environments).
7. **Characteristic vocabulary:** "Burden," "surveillance," "stewardship," "pipeline," "priority pathogens," "evidence-based," "intervention," "counterfactual," "DALYs," "attributable mortality."
8. **Temporal orientation:** Primarily forward-prescriptive ("what must be done") with a backward-causal evidence base ("what the data show"). Limited lateral-hermeneutic capacity.

Grammar Boundary: Grammar A would transform into Grammar B if its primary analytical variable shifted from pathogen biology to market structure — if it began treating pharmaceutical economics not as context but as the causal engine.

Grammar B: The Political Economy Grammar

Source Tradition: Health economics, development economics, global health governance scholarship, the O'Neill Review tradition, science and technology policy studies. Also the self-narration of pharmaceutical companies in their public advocacy for pull incentives.

Required Parameters:

1. **Primary analytical variable:** Market structure and economic incentive as the causal engine of the AMR crisis. The biological facts are taken as given; what drives the crisis is the failure of economic systems to produce appropriate responses.
2. **Signal selection rules:** Admits economic signals (R&D investment, pipeline economics, company bankruptcies, pricing models), governance signals (NAP implementation rates, UN declarations, trade data), and epidemiological signals reframed as economic externalities.
3. **Weighting priority:** Market failure indicators are weighted highest: industry disinvestment, the broken business model, the gap between social value and commercial return. Mortality data is admitted but reframed as the *consequence* of market failure rather than the primary signal.
4. **Causal chain type:** Structural-systemic. The causal chain runs: misaligned incentives → underinvestment in R&D → thin pipeline → inadequate therapeutic options → preventable deaths. The chain extends to: global trade structures → regulatory arbitrage in agricultural antibiotics → externalised resistance costs.
5. **Admissible closure:** The problem requires structural economic reform: pull incentives (subscription models, transferable exclusivity vouchers), push funding (public investment in early-stage R&D), and global coordination to prevent free-riding. Closure takes the form of "the correct incentive architecture will produce the correct outcomes."

Recommended Parameters:

6. **Excluded signals:** Tends to exclude or underweight ecological complexity, evolutionary dynamics that may outpace any incentive-driven pipeline, environmental reservoirs of resistance, and the epistemic limits of economic modelling when applied to biological systems.
7. **Characteristic vocabulary:** "Market failure," "pull incentive," "push funding," "delinked payment," "subscription model," "externality," "broken market," "return on investment," "public goods," "free-riding."
8. **Temporal orientation:** Forward-prescriptive (designing incentive architectures) with a lateral analytical component (comparing governance models across countries).

Grammar Boundary: Grammar B would transform into Grammar A if it began treating the market as mere context and the biology as causally fundamental. It would transform into Grammar C if it began questioning whether pharmaceutical products are the appropriate unit of intervention at all.

Grammar C: The Ecological-Systems Grammar

Source Tradition: One Health science, environmental microbiology, planetary health, veterinary public health, ecological theory, and increasingly, science and technology studies (STS) and multispecies ethnography. Intellectually rooted in systems ecology and complexity theory.

Required Parameters:

1. **Primary analytical variable:** The resistome — the total ecology of resistance genes, selection pressures, and transmission pathways across human, animal, and environmental domains — as an emergent property of industrial modernity's relationship to microbial life.
2. **Signal selection rules:** Admits signals from across the One Health spectrum: clinical resistance data, agricultural antibiotic use, environmental resistance gene prevalence, trade flows of antibiotics, wastewater data, wildlife carriage data, soil resistome studies. Privileges signals that demonstrate cross-domain transmission.
3. **Weighting priority:** Signals demonstrating ecological interconnection are weighted highest. The colistin/mcr-1 case is paradigmatic: its movement from agricultural use through gene transfer to human clinical resistance via environmental intermediaries demonstrates the grammar's core thesis.
4. **Causal chain type:** Non-linear, emergent, feedback-driven. Resistance is not caused by any single practice but emerges from the aggregate selective pressure exerted by industrial civilisation's antibiotic use across all domains simultaneously. Feedback loops (environmental reservoirs re-seeding clinical resistance) make linear intervention logic inadequate.
5. **Admissible closure:** The problem requires systemic transformation of humanity's relationship to microbial ecologies. Partial closure through specific interventions is possible but always provisional, because the underlying ecological dynamics are not addressable through discrete pharmaceutical or regulatory interventions alone. Closure often takes the form of "the problem as framed is itself the problem."

Recommended Parameters:

6. **Excluded signals:** Tends to exclude or underweight the demonstrated success of targeted interventions (paediatric mortality decline, colistin ban effects), the practical necessity of new drugs for patients dying now, and the institutional realities that constrain policy to implementable steps.
7. **Characteristic vocabulary:** "Resistome," "One Health," "multispecies," "emergence," "complexity," "feedback loop," "planetary health," "selective pressure," "environmental reservoir," "industrial ecology," "anthropocene."
8. **Temporal orientation:** Primarily lateral-hermeneutic (what does AMR *mean* about modernity?) with a deep backward-causal component (how did industrial antibiotic use create this ecological situation?). Less prescriptive than the other grammars, or prescriptive at a different scale.

Grammar Boundary: Grammar C would transform into Grammar A if it narrowed its unit of analysis from the resistome to the pathogen-drug combination. It would transform into Grammar B if it began treating the ecological analysis as instrumentally useful for designing better incentive structures rather than as an end in itself.

SECTION 4: SIGNAL PROCESSING

Grammar A Processing: The Biomedical-Institutional Grammar

The core fact is grim and precisely quantified: 1.27 million deaths directly attributable to bacterial antimicrobial resistance in 2019, with 4.95 million associated deaths, placing AMR above HIV/AIDS and malaria as a global cause of mortality. The GRAM project's 2024 update extends this into historical trend and forecast: more than one million people died from AMR each year between 1990 and 2021, and under the reference scenario, directly attributable deaths are projected to reach 1.91 million per year by 2050 — an increase of nearly 70% from 2022. Over the next quarter century, the cumulative toll could exceed 39 million lives.

These numbers anchor the grammar's processing. The six leading pathogens — *E. coli, S. aureus, K. pneumoniae, S. pneumoniae, A. baumannii*, and *P. aeruginosa* — account for 929,000 attributable deaths. The pathogen-drug specificity matters: MRSA alone caused over 100,000 attributable deaths in 2019 and more than 130,000 by 2021, more than doubling since 1990. Carbapenem resistance in Gram-negative bacteria is increasing faster than any other resistance type. This level of granularity is not incidental to the grammar — it *is* the grammar. The Biomedical-Institutional Grammar operates by decomposing the aggregate problem into specific pathogen-drug pairs, each amenable to targeted intervention.

The age-distribution shift is processed as evidence that intervention works where it is applied. AMR deaths in children under five fell by 50% between 1990 and 2021 — a consequence of improved paediatric infection prevention, vaccination (particularly pneumococcal), and neonatal care in many settings. This counter-signal (CS2) is absorbed comfortably: it confirms that evidence-based clinical and public health measures reduce AMR mortality. But the same data set reveals the problem's forward trajectory: deaths in those over 70 increased by more than 80% over the same period, driven by ageing populations, rising comorbidity, immunodeficiency, and the increasing prevalence of Gram-negative infections in elderly care settings. By 2050, the over-70s are projected to account for roughly two-thirds of all attributable AMR deaths.

The pipeline data is processed with calibrated alarm. Only 32 antibiotics are in clinical development targeting priority pathogens. Only 12 are classified as innovative. Only 4 target critical-priority pathogens. The grammar acknowledges the structural thinness of this pipeline but reads it primarily as a scientific challenge: the difficulty of discovering genuinely novel mechanisms of action against Gram-negative bacteria, which possess intrinsic resistance mechanisms that make them formidable targets. The 2024–2025 first-in-class approvals (gepotidacin, and the promising zosurabalpin programme against *A. baumannii*) are absorbed as evidence that the scientific problem is not insoluble — counter-signal CS3 integrates here — but the pace is inadequate relative to the evolutionary pace of resistance.

Agricultural antibiotic use is admitted but weighted as a secondary driver. The grammar accepts the One Health framework's premise that resistance crosses species boundaries. The colistin case is instructive: the mcr-1 gene's trajectory from Chinese pigs through environmental intermediaries to global human clinical isolates is acknowledged as a canonical example of agricultural-to-human transmission. But the grammar notes that the *largest* drivers of human clinical resistance remain hospital prescribing, community antibiotic overuse, and inadequate infection prevention and control — particularly in low- and middle-income settings where the AMR burden is highest. The contested signal on agricultural transmission (Contested Signal 1) is processed by admitting the connection while subordinating it to clinical drivers.

Surveillance is processed as both a current strength and a critical gap. The grammar's entire evidentiary architecture depends on laboratory data — isolate testing, resistance profiling, clinical outcome tracking. The persistent weakness of surveillance infrastructure in sub-Saharan Africa, which bears the highest per-capita AMR mortality rate, is therefore not merely a practical limitation but an epistemic one: the grammar cannot fully operate in settings where its primary data source does not exist. This is a genuine internal tension. The GRAM modelling addresses it through statistical extrapolation, but the grammar's own standards of evidence would normally resist such modelling in other contexts.

Counter-signal tension: Counter-signal CS5 — the persistence of resistance genes in antibiotic-free environments — creates real strain. If resistance persists without ongoing selective pressure, then the grammar's core causal logic (reduce exposure → reduce resistance) is incomplete. The grammar absorbs this by extending its causal model: resistance gene persistence requires ongoing surveillance and containment, not just stewardship. But the absorption is not clean; it begins to open into ecological territory.

Counterpoint within the grammar: A minority position within the Biomedical-Institutional community argues that the grammar's focus on new drug development is a category error — that the vast majority of AMR deaths globally would be prevented not by novel antibiotics but by universal access to existing ones, combined with clean water, sanitation, and basic infection prevention. The GRAM "better care" scenario

estimates 92 million cumulative deaths averted through these measures, compared to 11.1 million from a new Gram-negative drug pipeline. This internal dissent challenges the grammar's tendency to foreground the pipeline while backgrounding infrastructure.

Closure: Antimicrobial resistance is a measurable, rising cause of global mortality that requires a coordinated evidence-based response: expanded surveillance, strengthened stewardship, improved infection prevention, accelerated development of novel therapeutics, and universal access to existing antibiotics. The tools exist; the challenge is implementation.

This closure is a product of the processing architecture. The Biomedical-Institutional Grammar produces actionable, quantified, intervention-oriented conclusions because its five parameters — pathogen-level analysis, evidence hierarchies, mortality weighting, linear causation, and prescriptive closure — are designed to produce exactly this.

Grammar B Processing: The Political Economy Grammar

The same 1.27 million deaths enter this grammar not as a biological fact requiring clinical response but as the downstream consequence of a structural economic failure. The question is not "how do we treat resistant infections?" but "why does the world lack the tools to treat them, given that the scientific knowledge to develop those tools largely exists?"

The answer, this grammar argues, is legible in a single comparison: in 2020, the pharmaceutical industry invested £5.4 billion in oncology R&D and £125 million in antibiotics. That ratio — roughly 43:1 — is not a reflection of relative disease burden (AMR kills more people than many cancers) but of relative commercial return. Oncology drugs can be priced at hundreds of thousands of pounds per patient-year and prescribed continuously. Antibiotics must be used sparingly to preserve efficacy, cannot command high prices in most markets, and face generic competition rapidly. The rational profit-maximising firm exits antibiotics and enters oncology. This is not a failure of individual companies but of the incentive architecture of pharmaceutical capitalism.

The signal of industry disinvestment (S3) is therefore not context but the central causal mechanism. The fact that antibiotic development is "almost entirely absent from the 2025 clinical pipelines of the 20 largest pharma companies" is the headline, not the mortality count. The bankruptcies of antibiotic-focused companies — Achaogen, Melinta, Tetraphase — despite holding FDA-approved products are processed as proof that the market is structurally broken: companies that do the right thing are punished by the market for doing so. The pipeline's thinness (S2) is not primarily a scientific problem but a capital-allocation problem.

The UK subscription model (S6) is processed as the most important signal in the set — the first serious attempt by any government to address the fundamental incentive misalignment. By delinking payment from volume, the NHS has created a structure in which companies receive guaranteed revenue regardless of how many doses are prescribed, aligning profit with stewardship rather than sales. The pilot's design — £10 million per year per product, expanded to £5–20 million bands — is evidence of proof-of-concept. The August 2024 tender, estimating £1.9 billion over 16 years, represents a dramatic scaling of commitment.

But counter-signal tension operates here. The Political Economy Grammar must reckon with the scale problem: even £20 million per year per product in a single national market is inadequate to incentivise the billion-pound investments required for genuinely novel antibiotic development. The US PASTEUR Act remains stalled in the Senate. The EU's proposed transferable exclusivity voucher is under consideration but unimplemented. Sweden's pilot is small. The grammar's own logic implies that only coordinated G7/G20 action at the scale of hundreds of millions of additional R&D dollars per year — the WHO/G7 estimate of $250–400 million in additional annual push funding — would be sufficient. The gap between what the

grammar prescribes and what political economies have delivered is itself a signal that the grammar must process: *why* has collective action failed?

Agricultural antibiotic use (S4) is processed through the lens of regulatory arbitrage and trade. The colistin case (S5) is paradigmatic: China's 2017 ban is acknowledged as a regulatory success (counter-signal CS1), but the grammar immediately notes that between 2018 and 2021, China exported 2,664 tonnes of colistin to 21 countries, including to nations that had nominally banned agricultural colistin use (Bangladesh, Nigeria). The problem is not the regulation but the enforcement infrastructure, the global trade system, and the structural incentives facing farmers in low- and middle-income countries where intensive livestock production is expanding to meet rising protein demand. The political economy of meat production — particularly in South and Southeast Asia, where antibiotic use is projected to increase most steeply — is processed as inseparable from the political economy of AMR.

The National Action Plan data (S7) is processed sceptically. 178 countries have adopted NAPs, but this grammar treats plan adoption without dedicated funding or implementation mechanisms as performative compliance — a governance signal rather than an intervention signal. The proliferation of declarations (WHO Global Action Plan 2015, UN High-Level Meeting 2016, revised declaration 2024) is read as evidence of the gap between multilateral rhetoric and national-level structural reform.

Counter-signal tension: Counter-signal CS3 — the 2024–2025 first-in-class approvals — creates tension for the strongest version of the market-failure thesis. If GSK and Roche are developing genuinely novel antibiotics, is the market really broken? The grammar absorbs this by noting that GSK's gepotidacin programme and Roche's zosurabalpin were both substantially supported by public funding and academic discovery (the latter originated from Harvard research). The companies are executing development, but the risk was substantially socialised. This absorption works, but it modifies the closure: the market is not completely broken — it is broken for fully private investment, which means the correct intervention is public de-risking, not market replacement.

Counterpoint within the grammar: A dissenting position argues that pull incentives like the subscription model risk becoming a subsidy for pharmaceutical incumbents without addressing deeper structural issues — particularly global access. A product developed with pull-incentive support in the UK or US may remain inaccessible in sub-Saharan Africa, where the AMR burden is highest, due to pricing, supply chain, and regulatory barriers. The O'Neill Review's access agenda has been consistently subordinated to the innovation agenda in subsequent policy development.

Closure: Antimicrobial resistance is primarily a consequence of market failure in pharmaceutical innovation, compounded by governance failure in global coordination. The solution requires restructuring economic incentives — subscription models, pull incentives, push funding — at sufficient scale to make antibiotic development commercially viable, alongside enforceable international regulation of agricultural antibiotic use and equitable global access frameworks.

This closure is a product of the processing architecture. The Political Economy Grammar produces structural-reform conclusions because its parameters — market analysis, economic signal weighting, systemic causation, and incentive-design closure — are designed to produce exactly this.

Grammar C Processing: The Ecological-Systems Grammar

This grammar begins by refusing the premise that AMR is a "problem" to be "solved." It is, instead, an emergent property of industrial civilisation's relationship to the microbial world — a relationship characterised by the assumption that biological systems can be controlled through chemical intervention without ecological consequence.

The 1.27 million attributable deaths and 4.95 million associated deaths are admitted, but the grammar immediately notes what these numbers obscure: the vast, unmeasured ecology of resistance that operates below the threshold of clinical detection, in soils, waterways, wastewater treatment plants, wild animal populations, and the environmental microbiome. The clinical mortality data is the visible fraction of a phenomenon whose true scale is ecological, not epidemiological. The surveillance gap (S9) is therefore not merely a data limitation — as the Biomedical Grammar processes it — or a governance failure — as the Political Economy Grammar reads it — but a symptom of the epistemological narrowing that treats AMR as a clinical event rather than an ecological condition.

Agricultural antibiotic use (S4) is weighted as the central signal. The approximately 110,777 tonnes of veterinary antimicrobials used in 2019, projected to reach 143,481 tonnes by 2040, represents an ecological intervention of extraordinary scale — the sustained application of selective pressure to microbial populations across the planet's food production systems. The grammar processes this not as a regulatory problem but as an ecological fact: this volume of antibiotic application constitutes a planetary-scale experiment in directed evolution. The global commitment to 30–50% reduction by 2030 is processed against the projection of 29.5% increase under business-as-usual, and the grammar reads this gap as evidence not of inadequate political will (the Political Economy reading) but of the structural impossibility of significantly reducing antibiotic use without transforming the industrial model of food production itself.

The colistin tracer (S5) is processed as the grammar's demonstration case. The mcr-1 gene's journey is paradigmatic: originating in agricultural settings in China, moving through plasmid-mediated horizontal gene transfer across bacterial species, disseminating through global livestock trade, persisting in environmental reservoirs (wild bird populations carrying MCR-1 at 25% prevalence in Pakistani studies), contaminating waterways through manure runoff, and appearing in human clinical isolates on multiple continents. This is not a linear causal chain from farm to patient. It is an ecological web in which the human clinical encounter is one node among many. The grammar notes that China's colistin ban (CS1) achieved measurable reduction in mcr-1 prevalence domestically — a genuine counter-signal — but simultaneously, the gene continued spreading globally through trade, environmental persistence, and cross-species transmission. The ecology moves faster than the regulation.

Counter-signal CS5 — the persistence of resistance genes in antibiotic-free environments — is not a counter-signal within this grammar but a core confirming signal. Of course resistance persists: genes do not respect the on/off logic of regulatory intervention. Once a resistance gene has entered the environmental resistome, it becomes part of the microbial commons, subject to selection, drift, and horizontal transfer on evolutionary timescales. This is why the grammar is sceptical of stewardship as a sufficient response: stewardship addresses the rate of new resistance generation but not the stock of existing resistance in the environment.

The drug pipeline (S2) is processed differently from both other grammars. Rather than treating it as either a scientific challenge or a market failure, this grammar asks a prior question: is the correct response to an ecological disruption a new pharmaceutical product? Each new antibiotic introduced into clinical use will face the same evolutionary dynamics that degraded its predecessors. The grammar does not dismiss the immediate clinical need — people are dying of resistant infections and need treatment — but it treats the pipeline as a holding action, not a solution. The deeper intervention, in this grammar, is systemic: transforming agricultural practices, redesigning healthcare to minimise infection rather than treat it, managing wastewater and environmental antibiotic contamination, and fundamentally rethinking the assumption that chemical control of microbial populations is sustainable at industrial scale.

The One Health framework (S12) is processed with ambivalence. On one hand, it represents the grammar's victory at the level of rhetoric: the institutional world has adopted the language of cross-domain interconnection. On the other hand, the grammar notes that One Health as operationalised by the WHO/FAO/WOAH Quadripartite remains largely additive — human health plus animal health plus environmental health — rather than genuinely ecological. The Quadripartite structure replicates the very

jurisdictional boundaries that the ecological grammar seeks to dissolve. The environmental component, added late (UNEP joined in 2022), remains the weakest of the four.

Counter-signal tension: Counter-signal CS4 — Norway's elimination of antibiotics from salmon aquaculture while growing the industry — is the most threatening counter-signal for Grammar C's stronger claims. If one sector in one country could eliminate antibiotic use without ecological collapse, the grammar's pessimism about the impossibility of reform within existing structures is challenged. The grammar absorbs this by noting that Norwegian aquaculture reform occurred within a wealthy, well-regulated, technologically advanced, low-corruption governance context — conditions that do not obtain in the settings where agricultural antibiotic use is growing fastest. The absorption works but narrows the grammar's scope: its strongest claims apply to the global system, not to every local case.

Counterpoint within the grammar: A pragmatist wing argues that the ecological grammar's tendency toward systemic critique — "the problem as framed is itself the problem" — can produce analytical paralysis. Patients are dying now; farmers in LMICs depend on antibiotics now; environmental reservoirs exist now. The grammar's insistence on systemic transformation, however intellectually correct, may foreclose the incremental interventions that would save lives in the interim.

Closure: Antimicrobial resistance is an emergent ecological condition produced by industrial modernity's chemical assault on microbial ecosystems across human, animal, and environmental domains simultaneously. Pharmaceutical solutions are holding actions; regulatory interventions are geographically bounded; economic incentives address symptoms rather than causes. The adequate response requires transformation of the agricultural, healthcare, and environmental systems that generate sustained selective pressure — a civilisational challenge, not a policy problem.

This closure is a product of the processing architecture. The Ecological-Systems Grammar produces systemic-transformation conclusions because its parameters — ecological primary variable, cross-domain signal selection, interconnection weighting, non-linear causation, and provisional closure — are designed to produce exactly this.

SECTION 5: CLOSURE COMPARISON

i. Closure content

Grammar A closes on *implementation*: the tools exist, the evidence is clear, the task is to deploy stewardship, surveillance, infection prevention, and new therapeutics at scale. Grammar B closes on *structural reform*: the incentive architecture is broken and must be redesigned through subscription models, push funding, and global coordination. Grammar C closes on *systemic transformation*: the problem exceeds the scale of any policy intervention and requires fundamental change in industrial modernity's relationship to microbial ecology.

These closures are not simply different recommendations. They define different problems, such that success in one grammar's terms would not necessarily register as success in another's. Grammar A's ideal outcome — a robust pipeline and declining mortality — could coexist with Grammar C's nightmare scenario — a continuously escalating ecological arms race masked by pharmaceutical innovation.

ii. Signal divergence

All three grammars admit the GRAM mortality data (S1), but weight it differently: for A, it is the primary signal; for B, it is the consequence of the primary signal (market failure); for C, it is the visible surface of a deeper phenomenon. The sharpest divergence is on agricultural antibiotic use (S4): secondary for A, a trade-and-regulation problem for B, and the central ecological signal for C. The pipeline data (S2) produces

perhaps the most revealing divergence: A reads it as scientific challenge, B as market failure, and C questions whether the pipeline is the right category of response at all.

iii. Exclusion patterns

Grammar A systematically excludes structural economic analysis and ecological complexity. Grammar B excludes evolutionary dynamics that might outpace any incentive structure and environmental resistance reservoirs. Grammar C excludes the practical urgency of patients dying now and the demonstrated success of targeted interventions. Each grammar's exclusions are the other grammars' primary material.

iv. Causal architecture

Grammar A: linear-proximate (exposure → resistance → mortality → intervention). Grammar B: structural-systemic (incentive misalignment → underinvestment → thin pipeline → mortality). Grammar C: emergent-ecological (industrial selective pressure → resistome evolution → feedback loops → systemic condition). These are not merely different lengths of causal chain but different *shapes*: a line, a feedback loop with a chokepoint, and a web.

v. Closure range

Grammar A has the narrowest closure range: specific, quantified, intervention-targeted. Grammar B has a medium range: structural but still within the framework of policy reform. Grammar C has the widest range but the least operational specificity — "civilisational transformation" is not a policy brief.

vi. Closure timing

Grammar A reaches closure fastest and with most confidence: the data is clear, the interventions are known, the task is implementation. Grammar B reaches closure with moderate speed but less confidence, because its structural prescriptions (global coordination on pull incentives) depend on political conditions it cannot control. Grammar C resists closure by design, treating premature closure as part of the problem.

Ontological check: Do the three grammars constitute the same object? Not entirely. Grammar A constitutes AMR as a measurable disease burden amenable to clinical and public health intervention. Grammar B constitutes AMR as a market failure with health consequences. Grammar C constitutes AMR as an ecological condition of modernity. These share a name and overlap in the empirical data they reference, but they are substantially different objects. A patient dying of an MRSA bloodstream infection is the same event in all three grammars, but the *problem* that event instantiates is different: a treatment failure, a market failure, or an ecological inevitability. This ontological divergence is the deepest finding of the analysis.

SECTION 6: COUNTER-SIGNALS, LEAKAGE, AND STRESS TEST

Consolidated Counter-Signal Analysis

The five counter-signals operate differently across grammars:

CS1 (China colistin ban success) is most threatening to Grammar C, which tends toward pessimism about bounded interventions. Grammar C absorbs it by noting that the gene continued spreading globally despite domestic decline — but this absorption is partial.

CS2 (Under-5 mortality decline) is absorbed by all three grammars but most comfortably by Grammar A, which treats it as proof of concept for evidence-based intervention.

CS3 (First-in-class approvals 2024–2025) threatens Grammar B's strongest market-failure claims. Grammar B absorbs it by noting public funding involvement, but the absorption weakens the categorical force of "the market is broken."

CS4 (Norwegian salmon) threatens Grammar C's systemic pessimism. Grammar C absorbs it by limiting scope to non-LMIC contexts, a reasonable but constraining move.

CS5 (Resistance gene persistence) threatens Grammar A's stewardship logic. Grammar A absorbs it by extending its model to include environmental containment, but this opens the door to Grammar C's territory.

Leakage Instances

Grammar A into B: When Grammar A discusses the thin pipeline, it cannot avoid economic language — "commercially attractive," "return on investment," "broken market" appears even in NICE's own publications. The grammar leaks into political economy whenever it explains *why* the pipeline is thin rather than merely *that* it is thin.

Grammar B into C: When Grammar B discusses agricultural antibiotic use through trade data and regulatory arbitrage, it cannot fully contain the ecological implications. The colistin trade data — 2,664 tonnes exported from China to 21 countries — is an economic signal, but it describes an ecological intervention. Grammar B leaks into C whenever it follows the antibiotic molecule past the point of sale.

Grammar C into A: When Grammar C acknowledges that patients are dying now and that new drugs are needed as a holding action, it is operating Grammar A's logic. The pragmatist wing of the ecological grammar is Grammar A in ecological clothing.

Grammar Stress Test

Grammar A: Most threatening counter-signal — CS5 (Resistance gene persistence)

If resistance persists in environments without ongoing antibiotic pressure, then the grammar's foundational causal logic (reduce exposure → reduce resistance) is insufficient. Grammar A can absorb this by expanding its intervention logic to include environmental decontamination and gene surveillance, but this expansion fundamentally changes the grammar's scope — it begins to require ecological competence it does not possess. **Closure is conditional** on resistance gene persistence being a secondary rather than primary driver of clinical resistance. If environmental reservoirs prove to be a major re-seeding source for clinical resistance, Grammar A's stewardship-centred closure fails.

Grammar B: Most threatening counter-signal — CS3 (First-in-class approvals)

If pharmaceutical companies are developing genuinely novel antibiotics with public support, the "broken market" narrative requires significant qualification. Grammar B absorbs this by arguing that public de-risking is itself proof of market failure — but this absorption converts the grammar from a market-failure story to a public-investment story, which is a substantive transformation. **Closure is conditional** on the demonstrated insufficiency of current pull incentive scales. If the UK model scales internationally and the pipeline responds, Grammar B's closure succeeds but its strongest causal claims (the market is fundamentally broken) require revision to something softer (the market requires ongoing structural support).

Grammar C: Most threatening counter-signal — CS1 (China colistin ban success)

If a single country's regulatory intervention can measurably reduce resistance gene prevalence, then Grammar C's scepticism about bounded interventions is challenged. Grammar C absorbs this by noting the gene's continued global spread despite domestic decline, but this is an incomplete absorption: it concedes that national regulation can have national effects, which undermines the grammar's strongest claims about ecological ungovernabiilty. **Closure is conditional** on the global resistome operating as a genuinely interconnected system where local interventions have only local effects. If national actions prove to have significant national-level consequences, Grammar C's civilisational-transformation closure is too extreme.

SECTION 7: POST-ANALYSIS REFLEXIVE NOTE

Was my home grammar prediction correct?

Yes. My processing consistently gravitated toward Grammar A — specifically, toward treating the GRAM mortality data as the foundational signal and processing everything else in relation to it. I had to actively resist the pull to frame Grammar B and C as "perspectives on" the biomedical problem rather than as grammars that constitute different objects entirely.

Which grammar was actually hardest?

Grammar C was harder than Grammar B, contrary to my Step 0 prediction. Grammar B was challenging but remained within a framework I could operate — structural causation, policy prescription, institutional analysis. Grammar C required me to genuinely question whether the clinical encounter is the right level of analysis, which cuts against my deepest processing tendencies. The lateral-hermeneutic move — asking what AMR *signifies* rather than what should be *done* — was the most foreign operation.

Did I discover blind spots I did not predict?

Two. First, the ontological divergence was deeper than I expected. I anticipated that the grammars would process the same object differently; I discovered that they constitute substantially different objects. This is the difference between interpretive pluralism (multiple views of one thing) and ontological pluralism (multiple things sharing a name). Second, I underestimated how much Grammar A leaks into Grammar B. The biomedical community's own publications are saturated with economic framing — "broken market," "commercially unattractive" — to the point where it is unclear where Grammar A ends and Grammar B begins. The grammars are not as cleanly separated as the method's architecture implies.

How would the analysis differ from a different home grammar?

An analyst operating from Grammar C as home would likely have given more prominence to the environmental resistome data and less to the GRAM mortality estimates. They would probably have treated the UK subscription model not as a significant innovation but as a minor rearrangement of deck chairs. The entire signal set would have been different: wastewater resistance gene surveys, soil resistome studies, and wildlife carriage data would have been core signals rather than peripheral.

What does this exercise reveal about interpretive architectures?

That the deepest divergences in a "consensual" domain are not about disagreements over facts but about the prior question of what constitutes the relevant object of analysis. Everyone agrees AMR is serious. They do not agree about what "AMR" is — a clinical burden, a market failure, or an ecological condition. This prior constitution determines everything downstream: which signals are admitted, how they are weighted, what causal stories are constructible, and what closures are reachable. The method makes this visible with considerable precision.

SECTION 8: METHODOLOGICAL OBSERVATIONS

What worked well in v2.0?

The integration of counter-signals throughout processing rather than deferring them to a final section was a major improvement. It forced each grammar to reckon with disconfirming evidence *during* its own operation, producing more honest processing and revealing the mechanisms of absorption. The grammar boundary specification was surprisingly useful: defining the transformation conditions between grammars made the architecture of each grammar more precise.

The temporal orientation parameter proved to be one of the most diagnostic features. Grammar A's forward-prescriptive orientation, Grammar B's design orientation, and Grammar C's lateral-hermeneutic orientation are genuinely different cognitive operations, and naming them sharpened the analysis of each grammar's processing.

Where were the difficulties?

The principal difficulty was maintaining grammar fidelity during the processing essays. The temptation to evaluate from outside — to note that Grammar C "goes too far" or Grammar B "is too narrow" — was constant. The method requires the analyst to *operate* each grammar, not merely describe it, and this is genuinely difficult when one grammar's closure strikes the analyst as more reasonable than another's.

The 10–15 signal target was approximately right for this domain, but the signals themselves proved less discrete than the method assumes. The colistin case (S5) is simultaneously an epidemiological signal, a trade signal, and an ecological signal. The method's architecture treats signals as stable objects rotated through grammars; in practice, the signal *itself* changes shape under different grammars, which is an aspect the contested-signal framework partly captures but does not fully address.

Were the new features useful?

The **ontological check** was the single most productive addition. Without it, the analysis would have concluded that three grammars "interpret AMR differently." With it, the analysis reveals that they constitute different objects — a far more significant finding. The **stress test** was useful for making closure conditions explicit: knowing that each grammar's closure is conditional on a specific empirical claim prevents the analyst from treating any closure as definitive.

The **contested signals** framework was useful but could be refined. The current instruction asks what each signal *is* according to each grammar. A further refinement might ask how each grammar would *verify or falsify* the contested signal — i.e., what evidence would resolve the contestation within each grammar's own terms. This would more precisely locate the disagreement.

Suggestions for further refinement

First, the method might benefit from a **grammar interaction** section that explicitly maps how grammars influence each other over time. In the AMR case, Grammar B has substantially colonised Grammar A's vocabulary — the biomedical community now routinely speaks in economic terms. Tracking this migration would add a diachronic dimension to what is currently a synchronic analysis.

Second, the relationship between signal contestation and ontological divergence deserves formal specification. When a signal is "contested," is the disagreement about the signal's meaning (interpretive divergence) or about whether the signal *exists as that kind of thing* at all (ontological divergence)? The colistin case involves both, and the method would benefit from distinguishing them explicitly.

Third, the method might consider a **closure confidence rating** — a self-assessment by the analyst of how confidently each grammar reaches its closure, calibrated against the counter-signals it had to absorb. This would make the stress-test findings more comparable across grammars and across case studies.

End of analysis. Total word count: approximately 7,800 words.